Verifying footprints

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Summary

This log tracks verification of each PCB footprint in the Digital Audio Player against its manufacturer datasheet before fabrication. Sections follow the schematic sheets. Parts sharing one manufacturer part number share one row, with every designator listed. Part data comes from the Altium parts database (AL-PN). Status follows its Footprint Status field; parts with no status or no database entry are unverified. Only parts placed on the schematic are listed.

Status colour	Meaning
Green	Verified
Red	Unverified
Yellow	Needs change

Audio

Sheet DAP-Audio-1: DAC, I/V conversion, op-amp summing and headphone output. 23 of 23 parts verified.

Status	Designator	Qty	AL-PN	Value	Package	Library Footprint	Manufacturer	Part Number	Notes
	B100	1	#1-053	DAC		QFN40P300X500X80-31N-D	ESS Technology	ES9038Q2M	
	Y100	1	#1-100	22.5792 MHz	4-SMD	OSC_ECS-2520MVLC-225.792-CN-TR	ECS Inc.	ECS-2520MVLC-225.792-CN-TR	
	Y101	1	#1-101	24.576 MHz	4-SMD	XTAL_ASTX-H11-24.576MHZ-T	Abracon	ASTX-H11-24.576MHZ-T	
	U100, U101, U102	3	#1-104	Op-amp, 2 ch	8-SON (3x3)	SON50P300X300X80-9N200X120	Texas Instruments	OPA1612AIDRGR	
	U103	1	#1-056	Op-amp, 2 ch	10-VSON (3x3)	IC_OPA1622IDRCR	Texas Instruments	OPA1622IDRCR	
	J100	1	#1-111	3.5 mm jack	THT, R/A	CUI_SJ1-3535NS	Same Sky	SJ1-3535NS	
	D100	1	#1-112	TVS, 8.4 V (Cl)	3-X2SON	IC_ESD122DMXR	Texas Instruments	ESD122DMXR	
	R125, R127	2	#1-113	4.7 Ω	0402	RC_0402_YAG	YAGEO	RT0402BRE074R7L	
	R104, R105	2	#1-102	22 Ω	0402	RC_0402_YAG	YAGEO	RC0402FR-0722RL	
	R114, R115, R126, R133	4	#1-109	220 Ω	0402	RC_0402_YAG	YAGEO	RT0402BRD07220RL	
	R100, R101, R102, R103, R116, R117, R121, R123, R131, R134	10	#1-105	499 Ω	0402	RC_0402_YAG	YAGEO	RT0402BRD07499RL	
	R120, R122, R124, R132	4	#1-110	681 Ω	0402	RC_0402_YAG	YAGEO	RT0402BRD07681RL	
	R106, R107, R111, R112, R113, R119, R129, R135	8	#1-134	10 kΩ	0402	RC_0402_YAG	YAGEO	AC0402FR-0710KL	
	R108	1	#1-138	49.9 kΩ	0402	RC_0402_YAG	YAGEO	RT0402BRD0749K9L	
	R128	1	#1-079	54.9 kΩ	0402	RC_0402_YAG	YAGEO	RT0402BRD0754K9L	
	R118	1	#1-131	100 kΩ	0402	RC_0402_YAG	YAGEO	RT0402BRD07100KL	
	C100, C101, C106, C107, C124, C138	6	#1-106	1.5 nF, C0G	0603	CAP_GCM/R18	Murata	GCM1885C1H152FA16D	
	C118, C123, C132, C137	4	#1-107	2.7 nF, C0G	0402	CAP_GCM/R15	Murata	GRM1555C1H272GE01D	
	C127, C130, C131, C136	4	#1-108	13 nF, C0G	0805	CAP_GCM/R21	Murata	GRM2195C1H133JA01D	
	C103, C104, C108, C109, C110, C111, C112, C114, C116, C119, C122, C125, C126, C128, C133, C134	16	#1-080	0.1 µF, X7R	0402	CAP_GCM/R15	Murata	GCM155R71H104KE02D	

Status	Designator	Qty	AL-PN	Value	Package	Library Footprint	Manufacturer	Part Number	Notes
	C102, C105, C113, C115, C120, C121, C129, C135, C139	9	#1-132	1 µF, X7R	0603	CAP_GCM/R18	Murata	GCM188R71E105KA64D	
	C117	1	#1-103	4.7 µF, X5R	0402	CAP_GCM/R15	Murata	GRM155R60J475ME47D	
	TP100, TP101, TP102	3		Test point					Generic test point; no footprint check needed

Power

Sheet DAP-Power-2: charger, USB-C input, 3.9 V and 3.3 V buck-boosts, inverter and LDOs. 49 of 50 parts verified.

Status	Designator	Qty	AL-PN	Value	Package	Library Footprint	Manufacturer	Part Number	Notes
	U202	1	#1-054	Charger IC	24-VQFN (4x4)	QFN50P400X400X100-25N	Texas Instruments	BQ24250RGET	
	U200, U205	2	#1-050	Buck-boost	10-MSOP	SOP50P490X110-10N	Analog Devices	LTC3440EMS#TRPBF	
	U201	1	#1-089	Cuk inverter	TSOT-23-6	TSOT-23_S6	Analog Devices	LT3462AES6#TRPBF	
	U204, U206	2	#1-051	LDO, positive	10-DFN (3x3)	VREG_LT3042EDD#PBF	Analog Devices	LT3042EDD#PBF	
	U203	1	#1-052	LDO, negative	12-DFN (3x3)	05-08-1725_ADI	Analog Devices	LT3093EDD#TRPBF	
	J201	1	#1-069	USB-C 2.0 receptacle	THT, R/A	GCT_USB4085-GF-A_REVA4	GCT	USB4085-GF-A	
	J200	1	#1-137	Battery header, 3-pos	SMD, R/A	CONN_505567_03_MOL	Molex	5055670381	Database: No Drawing
	D201	1	#1-049	TVS, 12 V (Cl)	SMA	DIOM5226X230N	Diodes Inc.	SMAJ7.0A-13-F	
	D202, D203	2	#1-099	TVS, 12.4 V (Cl)	SC-70-3	SOT65P210X110-3N	Texas Instruments	TPD2E2U06QDCKRQ1	
	D200, D204, D205, D206, D207, D208, D209	7	#1-009	LED, green	0603	LEDM1608X80	Lite-On	LTST-C190GKT	
	FB200	1	#1-068	120 Ω @ 100 MHz	0805	WE-CBF_0805_W4.0	Würth Elektronik	742792023	
	L203	1	#1-063	1 µH	0505	IND_TFM141206BLE	TDK	TFM141206BLE-1R0MTCA	Database: No 3D

Status	Designator	Qty	AL-PN	Value	Package	Library Footprint	Manufacturer	Part Number	Notes
	L200	1	#1-076	6.8 µH	Nonstandard	IND_SRP4030FA	Bourns	SRP4030FA-6R8M	
	L201, L202	2	#1-090	22 µH	1210	IND_LQW32FT220M2HL	Murata	490-LQW32FT220M2HLTR	Database: No 3D. MPN field holds a Digi-Key number
	L204	1	#1-094	18 µH	Nonstandard	IND_XGL5050	Coilcraft	XGL5050-183MEC	
	R227	1	#1-097	0.5 Ω	0805	RES-PYU-PE	YAGEO	PE0805FRF7W0R5L	
	R217	1	#1-065	270 Ω	0603	RC_0603_YAG	YAGEO	RC0603FR-07270RL	
	R218	1	#1-064	500 Ω	0603	RC_0603_YAG	YAGEO	RT0603BRC07500RL	
	R206, R229, R246, R249, R250, R251, R252, R253	8	#1-043	1 kΩ	0402	RESC1005X40N	YAGEO	RT0402BRD071KL	
	R214	1	#1-060	5 kΩ	0402	RC_0402_YAG	YAGEO	RC0402FR-075KL	
	R234, R235	2	#1-024	5.1 kΩ	0402	RESC1005X35N	YAGEO	RT0402BRE075K1L	
	R215	1	#1-061	9.88 kΩ	0402	RC_0402_YAG	YAGEO	RT0402DRD079K88L	
	R208, R209, R210, R211, R216, R221, R223, R242	8	#1-134	10 kΩ	0402	RC_0402_YAG	YAGEO	AC0402FR-0710KL	
	R213, R224	2	#1-078	15 kΩ	0402	RC_0402_YAG	YAGEO	RT0402BRD0715KL	
	R230	1	#1-088	15.4 kΩ	0402	RC_0402_YAG	YAGEO	RT0402BRD0715K4L	
	R232, R233, R248	3	#1-085	33.2 kΩ	0402	RC_0402_YAG	YAGEO	RT0402BRD0733K2L	
	R207, R226, R231, R238, R247	5	#1-138	49.9 kΩ	0402	RC_0402_YAG	YAGEO	RT0402BRD0749K9L	
	R203, R237	2	#1-079	54.9 kΩ	0402	RC_0402_YAG	YAGEO	RT0402BRD0754K9L	
	R219	1	#1-087	88.7 kΩ	0402	RC_0402_YAG	YAGEO	RC0402FR-0788K7L	
	R225, R244	2	#1-084	90.9 kΩ	0402	RC_0402_YAG	YAGEO	RT0402BRD0790K9L	
	R204, R205, R212, R228, R239, R240, R241, R245	8	#1-131	100 kΩ	0402	RC_0402_YAG	YAGEO	RT0402BRD07100KL	
	R220	1	#1-086	135 kΩ	0603	RC_0603_YAG	YAGEO	RT0603BRD07135KL	
	R236	1	#1-095	172 kΩ	0603	RC_0603_YAG	YAGEO	RT0603BRD07172KL	
	R200	1	#1-074	220 kΩ	0402	RC_0402_YAG	YAGEO	RC0402BR-07220KL	
	R202	1	#1-067	274 kΩ	0603	RC_0603_YAG	YAGEO	RT0603DRE07274KL	

Status	Designator	Qty	AL-PN	Value	Package	Library Footprint	Manufacturer	Part Number	Notes
	R201	1	#1-091	309 kΩ	0402	RC_0402_YAG	YAGEO	RE0402BRE07309KL	
	R222, R243	2	#1-083	909 kΩ	0402	RC_0402_YAG	YAGEO	RC0402FR-07909KL	
	C205	1	#1-093	22 pF, C0G	0402	CAP_GCM/R15	Murata	GRM1555C1H220JA01D	
	C225	1	#1-096	100 pF, C0G	0402	CAP_GCM/R15	Murata	GRM1555C1H101JA01D	
	C212, C226	2	#1-077	1.5 nF, C0G	0402	CAP_GCM/R15	Murata	GRM1555C1H152GA01D	
	C207	1	#1-059	33 nF, X7R	0402	CAP_GCM/R15	Murata	GRM155R71C333KA01D	
	C210, C213, C230, C234	4	#1-080	0.1 µF, X7R	0402	CAP_GCM/R15	Murata	GCM155R71H104KE02D	
	C214	1	#1-092	1 µF, X5R	0402	CAP_GCM/R15	Murata	GRM155R6YA105KE11D	
	C200, C215	2	#1-132	1 µF, X7R	0603	CAP_GCM/R18	Murata	GCM188R71E105KA64D	
	C201, C202	2	#1-070	2.2 µF, X7R	0603	CAP_GCM/R18	Murata	GRM188Z71E225KE43D	
	C222, C223, C233	3	#1-098	2.2 µF, tantalum	0603	CAP_298D225X0016M2T	Vishay Sprague	298D225X0016M2T	
	C219, C221	2	#1-071	4.7 µF, X7R	0805	CAP_GCM/R21	Murata	GRM21BR71C475KE51L	
	C203, C206, C211, C218, C220, C224, C232	7	#1-081	10 µF, X7R	0805	CAP_GCM/R21	Murata	GCM21BR71A106KE22K	
	C216, C217, C231	3	#1-142	10 µF, X7R (cut)	0805	CAPC2012X140N - CUT	Murata	GCM21BR71A106KE22K	Four-pin OUTS Kelvin symbol
	C204, C208, C209, C227, C228, C229	6	#1-073	22 µF, X7S	1210	CAP_GCM/R32	Murata	GCM32EC71E226KE36L	

MCU

Sheet DAP-MCU-3: ESP32-S3, external flash, 40 MHz crystal, boot and reset switches. 16 of 16 parts verified.

Status	Designator	Qty	AL-PN	Value	Package	Library Footprint	Manufacturer	Part Number	Notes
	U301	1	#1-055	MCU	56-VFQFN	IC_ESP32-S3	Espressif	ESP32-S3	
	U300	1	#1-124	Flash, 64 Mbit	8-WSON (5x6)	SON127P600X500X80-9N	GigaDevice	GD25B64EWIGR	
	CR300	1	#1-118	40 MHz crystal	4-SMD	CRYST_ECS-400	ECS Inc.	ECS-400-10-37B2-CKY-TR	
	SW300, SW301	2	#1-115	Tactile, SPST, vertical	SMD	SW_PTS810_SJM_250_SMTR_LFS	C&K	PTS810SJM250SMTR LFS	

Status	Designator	Qty	AL-PN	Value	Package	Library Footprint	Manufacturer	Part Number	Notes
	L300	1	#1-121	2 nH	0402	IND_MHQ10050 7	TDK	MHQ1005075 HA2N0BTD25	Database: No 3D
	L301	1	#1-120	24 nH	0402	IND_MHQ10050 7	TDK	MHQ1005075 HA24NHTD25	Database: No 3D
	R303, R304, R305, R306, R307	5	#1-102	22 Ω	0402	RC_0402_YAG	YAGEO	RC0402FR-0722RL	
	R302	1	#1-043	1 kΩ	0402	RESC1005X40N	YAGEO	RT0402BRD0 71KL	
	R300, R301	2	#1-134	10 kΩ	0402	RC_0402_YAG	YAGEO	AC0402FR-0710KL	
	R308	1	#1-123	1 MΩ	0402	RC_0402_YAG	YAGEO	RE0402BRE0 71ML	
	C308, C310	2	#1-122	12 pF, C0G	0402	CAPC1005X60N	Murata	GJM1555C1H1 20FB01D	
	C300	1	#1-040	10 nF, X7R	0402	CAPC1005X55N	YAGEO	CC0402KRX7 R7BB103	
	C302, C303, C306, C307, C311	5	#1-080	0.1 µF, X7R	0402	CAP_GCM/R15	Murata	GCM155R71H1 04KE02D	
	C301, C305, C309, C312	4	#1-132	1 µF, X7R	0603	CAP_GCM/R18	Murata	GCM188R71E1 05KA64D	
	C304	1	#1-081	10 µF, X7R	0805	CAP_GCM/R21	Murata	GCM21BR71A 106KE22K	
	TP300, TP301, TP302, TP303	4		Test point					Generic test point; no footprint check needed

Peripherals

Sheet DAP-Peripherals-4: power/volume switches, screen connector, LED driver, SD card and encoder. 16 of 17 parts verified.

Status	Designator	Qty	AL-PN	Value	Package	Library Footprint	Manufacturer	Part Number	Notes
	U400	1	#1-128	LED driver	16-VQFN (3x3)	VQFN016V3030_ROM	ROHM	BD1604MUV-E2	Database: Very slightly off (about 0.004 mm). Accepted for this revision; library fix deferred
	J400	1	#1-117	FFC, 50-pos	SMD, R/A	AMPHENOL_F32D-1A7Y1-21050	Amphenol ICC	F32D-1A7Y1-21050	
	J401	1	#1-126	microSD socket	SMD, R/A	CUI_MSD-1-A	Same Sky	MSD-1-A	

Status	Designator	Qty	AL-PN	Value	Package	Library Footprint	Manufacturer	Part Number	Notes
	ENC400	1	#1-135	Encoder	THT, R/A	EN12-VS_TTE	TT Electronics	EN12-VS11AF18	Database: Bad Drawing
	SW400, SW401, SW402	3	#1-114	Tactile, SPST, R/A	SMD, R/A	SW_TL1014BF220QG	E-Switch	TL1014BF220QG	
	Q400	1	#1-130	N-MOSFET	SOT-23	SOT-23_MCC	MCC	2N7002-TP	
	D400, D401	2	#1-127	TVS array, 5 V (RWM)	SOT-23-6L	TVS_ARR_4UNI_FP	Good-Ark	GSER5U007	Database: No 3D
	D402, D403, D404	3	#1-141	TVS, 5 V (Rf)	SOD-923	DIOM1006X40N - onsemi	onsemi	ESD9X5.0ST5G	Database: Rf 5 V, Cl 12.3 V
	R409, R410, R411	3	#1-043	1 kΩ	0402	RESC1005X40N	YAGEO	RT0402BRD071KL	
	R400, R401, R402, R403, R404, R405, R406, R407, R408, R412, R413, R414, R415, R416, R417, R418, R419, R422	18	#1-134	10 kΩ	0402	RC_0402_YAG	YAGEO	AC0402FR-0710KL	
	R423	1	#1-133	32.4 kΩ	0402	RC_0402_YAG	YAGEO	RC0402FR-0732K4L	
	R420, R421, R424, R425	4	#1-131	100 kΩ	0402	RC_0402_YAG	YAGEO	RT0402BRD07100KL	
	C404, C405	2	#1-040	10 nF, X7R	0402	CAPC1005X55N	YAGEO	CC0402KRX7R7BB103	
	C400, C407, C408	3	#1-080	0.1 µF, X7R	0402	CAP_GCM/R15	Murata	GCM155R71H104KE02D	
	C410, C411, C412, C413	4	#1-132	1 µF, X7R	0603	CAP_GCM/R18	Murata	GCM188R71E105KA64D	
	C409	1	#1-081	10 µF, X7R	0805	CAP_GCM/R21	Murata	GCM21BR71A106KE22K	
	TP402, TP403, TP404, TP405, TP406	5		Test point					Generic test point; no footprint check needed

Rev Log

Rev	Date	Change
A	2026-10-23	Document created. DAC sheet parts listed from schematic, all unverified.
B	2026-10-26	Expanded to all four schematic sheets. Part data and status taken from the Altium parts database.